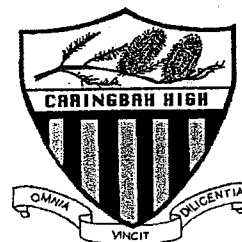


CARINGBAH HIGH SCHOOL

2010

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION



Mathematics Extension 1

General Instructions

Reading time - 5 minutes

Working time - 2 hours

Write using black or blue pen.

Board-approved calculators may be used.

A table of standard integrals is provided at the back of this paper.

Total marks - 84

Attempt Questions 1 - 7

All questions of equal value.

All necessary working should be shown in every question.

Start each question on a new page.

Question 1 (12 marks)**Marks**

(a) Factorise $2x^3 + 54$.

2

(b) Let $f(x) = e^x - 1$. What is the range of $f(x)$?

1

(c) Given that $\log_a b = 3 \cdot 4$ and $\log_a c = 4 \cdot 5$, find $\log_a \left(\frac{b}{c} \right)$.

1

(d) Differentiate $e^{3x} \sin x$

2

(e) Find the exact value of $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$.

3

(f) Using the substitution $u = x^2 + 1$, or otherwise, find the

3

exact value of $\int_0^3 \frac{2x}{x^2 + 1} \, dx$.

Question 2 (12 marks) Start a new page.

- (a) The function $f(x) = x^3 - \log_e(x+1)$ has a root near $x = 1$. **3**

Use one application of Newton's method to obtain another approximation to this root. Give your answer correct to two decimal places.

- (b) The polynomial $q(x) = x^3 + 3x^2 + ax + b$ has a factor of $(x-2)$ and a remainder of -9 when divided by $(x+1)$. **3**

Find the values of a and b .

- (c) (i) Express $\sqrt{3} \sin x - \cos x$ in the form $A \sin(x - \theta)$ where **2**
 $0 \leq \theta \leq \frac{\pi}{2}$.

- (ii) Hence, or otherwise, solve $\sqrt{3} \sin x - \cos x = 1$ **2**
for $0 \leq x \leq 2\pi$. Give your answer in exact form.

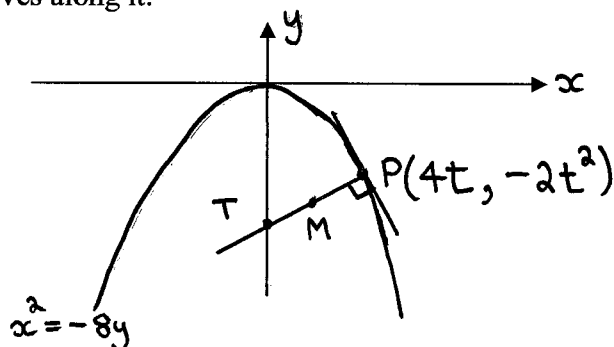
- (d) Evaluate $\int_0^{\ln 3} e^{2x} dx$. **2**

Question 3 (12 marks) Start a new page.

(a) Find $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}$

1

- (b) The diagram shows the parabola $x^2 = -8y$ and the point $P(4t, -2t^2)$ which moves along it.



- (i) Show that the equation of the normal at P is $y = \frac{x}{t} - 2t^2 - 4$. 2
- (ii) This normal meets the y -axis at T . The midpoint of PT is M . Find the coordinates of T and M . 2
- (iii) Find the Cartesian equation of the locus of M . 1
- (c) (i) On the same set of axes sketch the graphs of $y = \cos 2x$ and $y = -\frac{1}{2}x$ for $0 \leq x \leq \pi$. (Show at least 2 points which lie on $y = -\frac{1}{2}x$.) 2
- (ii) Use the graph to determine the number of solutions there are to the equation $\cos 2x = -\frac{1}{2}x$ for $0 \leq x \leq \pi$. 1
- (d) Use mathematical induction to prove that $7^n - 1$ is divisible by 3, for all integers $n \geq 1$. 3

Question 4 (12 marks) Start a new page.

- (a) A particle is moving in a straight line with its acceleration as a function of x given by $\ddot{x} = 3x^2$. It is initially 1 metre to the right of the origin and is travelling with a velocity of $-\sqrt{2}$ metres per second.

(i) Show that $\dot{x} = -\sqrt{2x^3}$ 2

(ii) Hence show that $x = \frac{2}{(t + \sqrt{2})^2}$ 3

- (b) (i) Find the x -coordinates of stationary points on the curve 3

$$y = x - 2 \sin x \quad \text{for } 0 \leq x \leq 2\pi.$$

- (ii) Write down the y -coordinates of these stationary points in exact form. 1

- (iii) Find the nature of each of the stationary points. 1

- (iv) Find the coordinates of those points on the curve corresponding to $x = 0$ and 2π . 1

- (v) Hence sketch the curve for $0 \leq x \leq 2\pi$. (You do not have to find any other intercepts with the axes or points of inflexion.) 1

Question 5 (12 marks) Start a new page.

- (a) The displacement x metres of a particle moving in simple harmonic motion is given by $x = 5 \sin(\pi t + \frac{\pi}{2})$ where the time t is in seconds.
- (i) What is the period of the oscillation? 1
- (ii) Show that the particle's oscillation starts 5 metres to the right of the equilibrium position. 1
- (iii) What is the speed of the particle as it moves through the equilibrium position? 2
- (iv) Show that the acceleration of the particle is proportional to the displacement from the equilibrium position. 2
- (b) Show that for $0 < x < 1$, $\frac{d}{dx} \left[\sin^{-1} \sqrt{1 - x^2} \right] = \frac{-1}{\sqrt{1 - x^2}}$ 3
- (c) (i) Sketch the graph of $y = \tan x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. 1
- (ii) By using (i), or otherwise, find those values of x 2
satisfying $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ for which the geometric series
 $1 + \sqrt{3} \tan x + 3 \tan^2 x + 3\sqrt{3} \tan^3 x + \dots$ has a limiting sum.

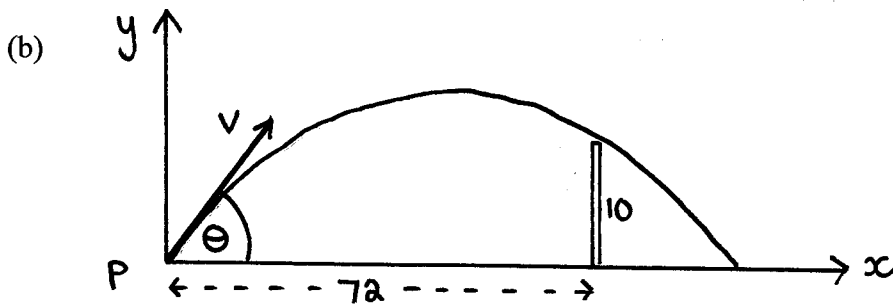
Question 6 (12 marks) Start a new page.

(a) Consider the function $f(x) = x^2 - 4x + 7$.

(i) Write down the largest positive domain for which $f(x)$ has an inverse function $f^{-1}(x)$. 1

(ii) Find the inverse function $f^{-1}(x)$. 2

(iii) Sketch $y = f^{-1}(x)$ and $y = f(x)$ on the same diagram. 1



A ball is hit with initial velocity v from a point P on the ground at an angle of θ with the horizontal.

(i) The equations of the motion of the ball are:

$$\ddot{x} = 0 \quad \text{and} \quad \ddot{y} = -10 \quad (\text{i.e. acceleration due to gravity is } 10\text{m/s}^2)$$

Using calculus, show that the position of the ball at time t is given by: 3

$$x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - 5t^2$$

(ii) After 2 seconds the ball just clears a wall which is 10 metres high and 72 metres from P . 3

Calculate the angle of projection θ (to the nearest degree) and the initial velocity v .

(iii) How far from P does the ball land? 2

Question 7 (12 marks) Start a new page.

(a) Find $\int \frac{1}{\sqrt{49 - x^2}} dx$.

1

- (b) Nicole is staying in the Blue Mountains. She takes a cup of coffee whose temperature is $80^\circ C$ outside where the air temperature is $5^\circ C$.

The temperature of the coffee, $T^\circ C$, after t minutes outside satisfies the equation

$$\frac{dT}{dt} = -k(T - 5).$$

- (i) Show that $T = 5 + 75e^{-kt}$ satisfies both the above equation and the initial condition.

2

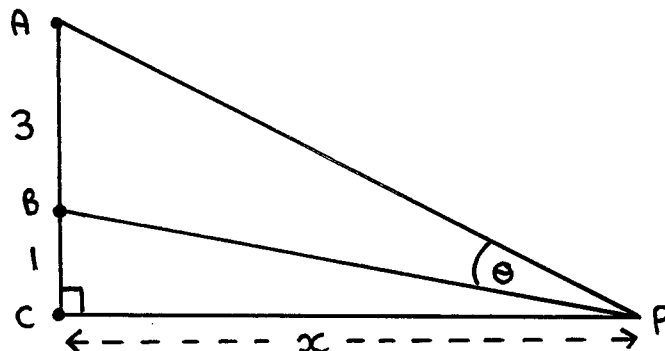
- (ii) After 10 minutes outside the temperature of the coffee is $57^\circ C$.
After how many minutes (to the nearest minute) will the coffee's temperature be $20^\circ C$?

3

Question 7 continues on page 9.

Question 7 (continued)

(c)



In the diagram, the goal posts A and B in a children's soccer game are 3 metres apart. Goal post B is 1 metre from the corner post C . A player stands at P on the boundary line x metres from C . The goal posts A and B subtend an angle of θ at P .

- (i) Show that $\theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$. 2
- (ii) Show that θ is a maximum when $x = 2$. 2
- (iii) Deduce that the maximum angle subtended at P is $\theta = \tan^{-1} \frac{3}{4}$. 2

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

CHS EXT. 1 TRIAL HSC 2010

1. a) $2(x^3 + 27) = 2(x+3)(x^2 - 3x + 9)$

b) $y > -1$

c) $\log_a\left(\frac{b}{c}\right) = \log_a b - \log_a c$
 $= 3 \cdot 4 - 4 \cdot 5$
 $= -1 \cdot 1$

d) $u = e^{3x}$ $v = \sin x$
 $u' = 3e^{3x}$ $v' = \cos x$

PRODUCT RULE: $y' = vu' + uv'$
 $= \sin x \cdot 3e^{3x} + e^{3x} \cdot \cos x$
 $= e^{3x}(3\sin x + \cos x)$

e) $\cos 2x = 2\cos^2 x - 1$

$\therefore \cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$\int_0^{\frac{\pi}{4}} \cos^2 x = \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 + \cos 2x \, dx$

$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$

$= \frac{1}{2} \left[\left(\frac{\pi}{4} + \frac{1}{2} \sin \frac{\pi}{2} \right) - \left(0 + \frac{1}{2} \sin 0 \right) \right]$

$= \frac{\pi}{8} + \frac{1}{4}$

f) $x=0: u=1$

$x=3: u=10$

$\frac{du}{dx} = 2x$

$\therefore du = 2x \, dx$

$\therefore \int_0^3 \frac{2x \, dx}{x^2 + 1} = \int_1^{10} \frac{du}{u}$
 $= [\ln u]_1^{10}$
 $= \ln 10 - \ln 1$
 $= \ln 10$

2. a) $f'(x) = 3x^2 - \frac{1}{x+1}$

$f'(1) = 3 - \frac{1}{2} = 2.5$

and $f(1) = 1 - \ln 2 \doteq 0.307$

$z_2 = z_1 - \frac{f(z_1)}{f'(z_1)}$ NEWTON'S METHOD

$= 1 - \frac{0.307}{2.5}$

$\doteq 0.88$

b) $(x-2)$ factor $\therefore q(2)=0$

i.e. $2a+b=-20$ ①

rem. = 9 when $\div$ by $(x+1) \therefore q(-1)=-9$

i.e. $-a+b=-11$ ②

Solving ① and ②: $a=-3, b=-14$

c) i) $\sqrt{(\sqrt{3})^2 + 1^2} = 2$

$\therefore \sqrt{3} \sin x - \cos x = 2 \left(\frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x \right)$

$= 2(\cos \theta \sin x - \sin \theta \cos x)$

$= 2 \sin(x-\theta)$ where $\cos \theta = \frac{\sqrt{3}}{2}$
 and $\sin \theta = \frac{1}{2}$
 i.e. $\theta = \frac{\pi}{6}$

$= 2 \sin\left(x - \frac{\pi}{6}\right)$

ii) $\sqrt{3} \sin x - \cos x = 1$

$2 \sin\left(x - \frac{\pi}{6}\right) = 1$

$\sin\left(x - \frac{\pi}{6}\right) = \frac{1}{2}$

$\left(x - \frac{\pi}{6}\right) = \frac{\pi}{6}, \frac{5\pi}{6}$

$x = \frac{\pi}{3}, \pi$

d) $\frac{1}{2} [e^{2x}]_0^{\ln 3} = \frac{1}{2} [e^{2 \ln 3} - e^0]$

$= \frac{1}{2} [e^{\ln 9} - 1]$

$= \frac{1}{2} [9 - 1]$

$= 4$

3. a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{5}{2} = 1 \times \frac{5}{2}$
 $= 2\frac{1}{2}$

$$3 \text{ (b) } y = -\frac{1}{8}x^2$$

$$y' = -\frac{1}{4}x$$

$$\text{At } (4t, -2t^2): m_{\text{TAN}} = -\frac{1}{4} \times 4t = -t$$

$$\therefore m_{\text{NORM}} = \frac{1}{t}$$

$$\text{EQUON NORM: } y + 2t^2 = \frac{1}{t}(x - 4t)$$

$$y = \frac{x}{t} - 2t^2 - 4$$

$$\text{(ii) } T(0, -2t^2 - 4)$$

$$M: x = \frac{4t+0}{2}, y = \frac{-2t^2 + -2t^2 - 4}{2}$$

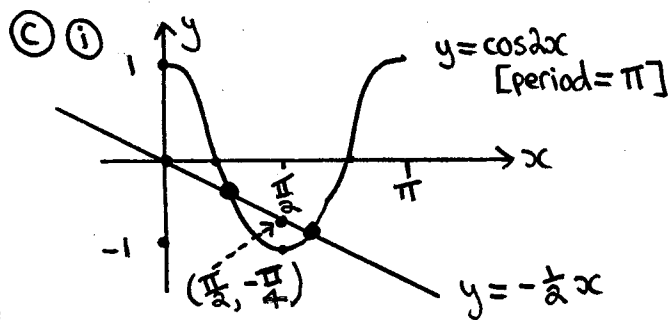
$$= 2t, \quad = -2t^2 - 2$$

$$\text{i.e. } M(2t, -2t^2 - 2)$$

$$\text{(ii) sub } t = \frac{x}{2} \text{ into } y = -2t^2 - 2$$

$$y = -2 \times \frac{x^2}{4} - 2$$

$$= -\frac{1}{2}x^2 - 2$$



$$\text{(ii) 2 points of intersection.}$$

$$\therefore \cos 2x = -\frac{1}{2}x \text{ has 2 solns}$$

$$\text{(d) STEP 1: Prove true for } n=1$$

$$7^1 - 1 = 6 \text{ which is divis. by 3}$$

$$\text{STEP 2: Assume true for } n=k$$

$$\text{i.e. } 7^k - 1 = 3M \text{ (M an integer)}$$

$$[\text{OR } 7^k = 3M + 1]$$

$$\text{Hence prove true for } n=k+1$$

$$\text{i.e. } 7^{k+1} - 1 \text{ also divis. by 3}$$

$$\text{Now } 7^{k+1} - 1 = 7 \cdot 7^k - 1$$

$$= 7 \cdot (3M + 1) - 1 \text{ by our assumption}$$

$$= 21M + 7 - 1$$

$$= 21M + 6$$

$$= 3(7M + 2) \text{ which is divis. by 3}$$

$$\text{i.e. if true for } n=k \text{ then true for } n=k+1$$

$$\text{STEP 3: we assumed true for } k \text{ and hence proved true for } n=k+1.$$

$$\text{Since true for } n=1 \text{ then by the Principle of Mathematical Induction true for all positive integers.}$$

$$4. \text{ (a) (i) } \frac{d}{dx}(\frac{1}{2}v^2) = 3x^2$$

$$\frac{1}{2}v^2 = x^3 + c$$

$$\text{Sub } x=1, v=-\sqrt{2}: 1 = 1 + c \Rightarrow c = 0$$

$$\frac{1}{2}v^2 = x^3$$

$$v^2 = 2x^3$$

$$v = \pm \sqrt{2x^3} \text{ but take -ve since } v = -\sqrt{2} \text{ initially}$$

$$v = -\sqrt{2x^3}$$

$$\text{(ii) } \frac{dx}{dt} = -\sqrt{2x^3}$$

$$\frac{dt}{dx} = -\frac{1}{\sqrt{2x^3}} = -\frac{1}{\sqrt{2}} x^{-3/2}$$

$$t = -\frac{1}{\sqrt{2}} \cdot 2x^{-1/2} + c$$

$$= \frac{\sqrt{2}}{\sqrt{x}} + c$$

$$\text{Sub } t=0, x=1: 0 = \sqrt{2} + c$$

$$c = -\sqrt{2}$$

$$\text{i.e. } t = \frac{\sqrt{2}}{\sqrt{x}} - \sqrt{2}$$

$$t + \sqrt{2} = \frac{\sqrt{2}}{\sqrt{x}}$$

$$(t + \sqrt{2})^2 = \frac{2}{x}$$

$$x = \frac{2}{(t + \sqrt{2})^2}$$

$$\text{(b) (i) } \frac{dy}{dx} = 1 - 2\cos x$$

$$\text{Stationary points } \frac{dy}{dx} = 0:$$

$$1 - 2\cos x = 0$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

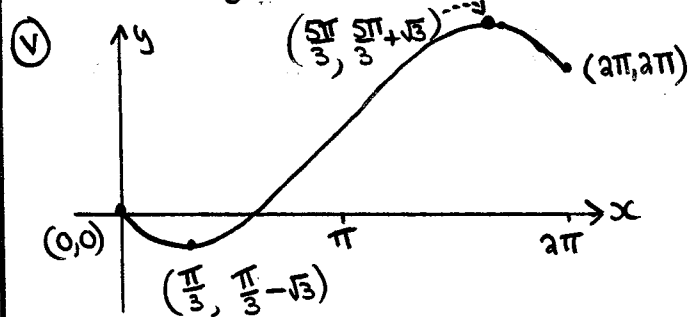
4. (b) (ii) $x = \frac{\pi}{3}$, $y = \frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2} = \frac{\pi}{3} - \sqrt{3}$
 $x = \frac{5\pi}{3}$, $y = \frac{5\pi}{3} - 2 \times -\frac{\sqrt{3}}{2} = \frac{5\pi}{3} + \sqrt{3}$

(iii) $\frac{d^2y}{dx^2} = 2 \sin x$

$(\frac{\pi}{3}, \frac{\pi}{3} - \sqrt{3})$: $\frac{d^2y}{dx^2} = 2 \sin \frac{\pi}{3} = \sqrt{3} > 0$
 $\therefore$ min. turning point

$(\frac{5\pi}{3}, \frac{5\pi}{3} + \sqrt{3})$: $\frac{d^2y}{dx^2} = 2 \sin \frac{5\pi}{3} = -\sqrt{3} < 0$
 $\therefore$ max. turning point

(iv) $x=0$: $y = 0 - 2 \sin 0 = 0$ i.e. $(0,0)$
 $x=2\pi$: $y = 2\pi - 2 \sin 2\pi = 2\pi$ i.e. $(2\pi, 2\pi)$



5. (a) (i) period = $\frac{2\pi}{n} = \frac{2\pi}{\pi} = 2s$

(ii) sub $t=0$: $x = 5 \sin \frac{\pi}{2}$
 $= 5 \times 1$
 $= 5$ i.e. 5m right of $x=0$.

(iii) when $x=0$: $5 \sin(\pi t + \frac{\pi}{2}) = 0$

$\pi t + \frac{\pi}{2} = 0, \pi, \dots$

$\pi t = -\frac{\pi}{2}, \frac{\pi}{2}, \dots$

$t = -\frac{1}{2}, \frac{1}{2}, \dots$ ignore $t = -\frac{1}{2}$

i.e. thru $x=0$ when $t = \frac{1}{2}s$

$v = \dot{x} = 5\pi \cos(\pi t + \frac{\pi}{2})$

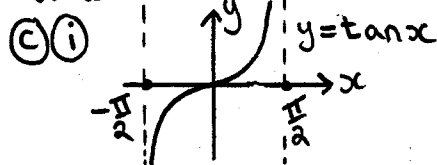
when $t = \frac{1}{2}$: $v = 5\pi \cos(\frac{\pi}{2} + \frac{\pi}{2})$
 $= 5\pi \cos \pi$
 $= -5\pi$ m/s

i.e. speed = 5π m/s

(iv) $a = \ddot{x} = -5\pi^2 \sin(\pi t + \frac{\pi}{2})$
 $= -\pi^2 \times 5 \sin(\pi t + \frac{\pi}{2})$
 $= -\pi^2 x$

i.e. acceleration proportional to displacement

(b) $\frac{1}{\sqrt{1-(\sqrt{1-x^2})^2}} \times \frac{-x}{\sqrt{1-x^2}}$
 $= \frac{1}{\sqrt{1-(1-x^2)}} \times \frac{-x}{\sqrt{1-x^2}}$
 $= \frac{1}{\sqrt{x^2}} \times \frac{-x}{\sqrt{1-x^2}}$
 $= \frac{-1}{\sqrt{1-x^2}}$
 since $\frac{d}{dx}(1-x^2)^{\frac{1}{2}} = \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot -2x = \frac{-x}{\sqrt{1-x^2}}$



(ii) series has limiting sum if $-1 < r < 1$.

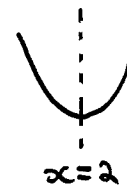
Here $r = \sqrt{3} \tan x$

i.e. $-1 < \sqrt{3} \tan x < 1$

$-\frac{1}{\sqrt{3}} < \tan x < \frac{1}{\sqrt{3}}$

$\therefore -\frac{\pi}{6} < x < \frac{\pi}{6}$

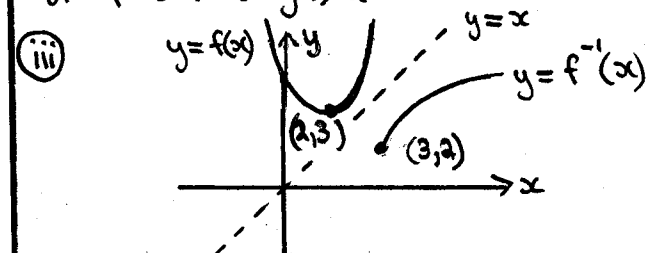
6. (a) (i) axis of symmetry: $x = -\frac{-4}{2} = 2$



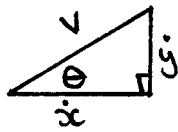
$\therefore$ DOMAIN $x \geq 2$

(ii) swap x, y : $x = y^2 - 4y + 7$
 $x - 7 = y^2 - 4y$
 $x - 7 + 4 = y^2 - 4y + 4$
 $x - 3 = (y - 2)^2$
 $y - 2 = \pm \sqrt{x - 3}$
 $y = 2 \pm \sqrt{x - 3}$

TAKE $y = 2 + \sqrt{x - 3}$ SINCE RANGE OF $f^{-1}(x)$ IS $y \geq 2$



6. (b) (i) INITIALLY



$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

HORIZONTAL

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$\therefore \dot{x} = v \cos \theta$$

$$x = vt \cos \theta + k$$

$$\text{sub } t=0, x=0 \Rightarrow k=0$$

$$x = vt \cos \theta$$

VERTICAL

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$\text{sub } t=0, \dot{y} = v \sin \theta$$

$$\Rightarrow c = v \sin \theta$$

$$\dot{y} = v \sin \theta - 10t$$

$$y = vt \sin \theta - 5t^2 + k$$

$$\text{sub } t=0, y=0 \Rightarrow k=0$$

$$y = vt \sin \theta - 5t^2$$

(ii) sub $t=2, x=72$ into $x = vt \cos \theta$

$$72 = 2v \cos \theta \quad (1)$$

sub $t=2, y=10$ into $y = vt \sin \theta - 5t^2$

$$10 = 2v \sin \theta - 20$$

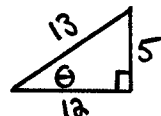
$$30 = 2v \sin \theta \quad (2)$$

$$(2) \div (1): \frac{2v \sin \theta}{2v \cos \theta} = \frac{30}{72}$$

$$\tan \theta = \frac{5}{12}$$

$$\theta \doteq 23^\circ$$

since $\tan \theta = \frac{5}{12} \therefore \cos \theta = \frac{12}{13}$



$$[\sin \theta = \frac{5}{13}]$$

so (1): $72 = 2v \times \frac{12}{13}$

$$v = 39 \text{ m/s}$$

(ii) $y = 39t \times \frac{5}{13} - 5t^2 = 15t - 5t^2$
hits ground when $y=0: 15t - 5t^2 = 0$
 $5t(3-t) = 0$
 $t = 0, 3.$

$$\therefore \text{dist. from P: } x = 39 \times 3 \times \frac{12}{13} = 108 \text{ m}$$

7. (a) $\sin^{-1} \frac{x}{7} + c$

(b) (i) $T = 5 + 75e^{-kt}$

$$\frac{dT}{dt} = -k \times 75e^{-kt}$$

$$= -k(T-5) \text{ since } T-5 = 75e^{-kt}$$

also when $t=0: T = 5 + 75e^0 = 80$

(ii) sub $t=10, T=57: 57 = 5 + 75e^{-10k}$

$$52 = 75e^{-10k}$$

$$\frac{52}{75} = e^{-10k}$$

$$k = \ln\left(\frac{52}{75}\right) \div -10$$

$$\doteq 0.0366$$

sub $T=20: 20 = 5 + 75e^{-0.0366t}$

$$15 = 75e^{-0.0366t}$$

$$0.2 = e^{-0.0366t}$$

$$t = \ln 0.2 \div -0.0366$$

$$\doteq 44 \text{ min}$$

(c) (i) Let $\angle BPC = \alpha$

In $\triangle ACP: \tan(\theta + \alpha) = \frac{4}{x}$

$$\therefore \theta + \alpha = \tan^{-1} \frac{4}{x}$$

In $\triangle BCP: \tan \alpha = \frac{1}{x}$

$$\therefore \alpha = \tan^{-1} \frac{1}{x}$$

Now $\theta = (\theta + \alpha) - \alpha = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$

(ii) $\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{4}{x}\right)^2} \times -\frac{4}{x^2} - \frac{1}{1 + \left(\frac{1}{x}\right)^2} \times -\frac{1}{x^2}$
$$= \frac{-4}{x^2 + 16} + \frac{1}{x^2 + 1}$$

when $x=2: \frac{d\theta}{dx} = \frac{-4}{20} + \frac{1}{5} = 0$

$\therefore$ stationary point when $x=2.$

$$\frac{d^2\theta}{dx^2} = \frac{8x}{(x^2 + 16)^2} - \frac{2x}{(x^2 + 1)^2}$$

when $x=2: \frac{d^2\theta}{dx^2} = \frac{16}{400} - \frac{4}{25} = -\frac{3}{25} < 0$

$\therefore \theta$ max. when $x=2.$

(iii) So max. $\theta = \tan^{-1} 2 - \tan^{-1} \frac{1}{2}$

let $x = \tan^{-1} 2 \therefore \tan x = 2$

and $y = \tan^{-1} \frac{1}{2} \therefore \tan y = \frac{1}{2}$

i.e. $\theta = x - y$

$$\tan \theta = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$= \frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}}$$

$$= \frac{3}{4}$$

$$\therefore \theta = \tan^{-1} \frac{3}{4}$$